

MLJ Contribution Information Sheet

Main claim and importance

Failure-Trace Allocator (FTA) is a gold-free, runtime-legal selective-deferral policy for matched-budget LLM reasoning. It improves answer selection by using logged failure-trace metadata and external consensus signals rather than adding more candidates or using gold labels at runtime.

This matters for machine learning because it studies post-generation budget allocation under a strict selection contract. The question is not just how to scale compute, but how to choose the best answer from existing model outputs once the budget has already been spent.

Evidence

- Final-300: $260/300 = 86.67\%$
- Aggregate-720: $581/720 = 80.69\%$
- Source-stratified bootstrap confidence intervals are positive versus L1, S1, TALE, and the source-local best external baseline.
- Against the pooled four-answer ensemble, FTA is ahead by point estimate but not statistically separated; the appropriate conclusion is competitiveness with strong aggregation baselines rather than broad superiority over ensembling.
- Non-retained ablation variants are reported as negative or insufficient evidence.

Reproducibility package

The submission package is designed to support offline reproduction of the reported metrics and paper tables from processed artifacts, including per-example records, manifests, method summaries, bootstrap summaries, and no-API table-generation scripts. Full end-to-end live regeneration requires commercial provider credentials and may vary because of provider-side nondeterminism, model/version changes, rate limits, or transient service failures; no API keys, secrets, `.env` files, or private credentials are included.

Closest related work

The manuscript is related to chain-of-thought and self-consistency, test-time compute scaling, budget-forcing / s1-style methods, TALE-style prompt budgeting, tree/graph search, verifier-guided selection, and cost-aware routing. The difference is that this paper operates on runtime-legal frontier and external signals after the budget has already been spent, with no gold labels at decision time.

Related submission disclosure

The author will disclose the related conference manuscript under review. That manuscript addresses a broader frontier-search allocation formulation with tree-search policy controls and compute-budget surface analysis. This MLJ manuscript instead focuses on Failure-Trace Allocator (FTA),

GSM8K Final-300 / Aggregate-720, L1 / S1 / TALE / source-local best external comparisons, source-stratified bootstrap confidence intervals, and runtime-legal no-gold triggers.